Spherical Cyclic Time v1.6-F
Conditional FDT Closure Interpretation and Publication Update

Spherical Cyclic Time / Tempo Sferico Ciclico Project

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Abstract

This addendum closes the v1.6 conditional FDT branch as a toy/informational refinement. The branch strengthens the perturbative transfer kernel by replacing the earlier free power-law regeneration ansatz with a preregistered Model-A critical/massless fluctuation-dissipation closure. The result is a non-empty conditional toy restart region, but it is not a derivation of a physical cosmological restart. The bath, free-energy parameters, transition scale, and physical perturbation variable remain external to the current framework.

Claim boundary

This document is not peer reviewed. It makes no claim of a physical cyclic cosmology, no bounce model claim, no observational prediction, no microphysical derivation of a cosmological restart, and no derivation of a true covariant causal-patch entropy. It records a conditional toy closure only.

Conditional FDT closure

The preregistered v1.6-D closure was a critical/massless Model-A toy bath,

$$F[\delta] = \frac{1}{2}\int \frac{d^3k}{(2\pi)^3} c k^2 |\delta_k|^2, \tag{1}$$

with linear stochastic transition dynamics

$$\frac{d\delta_k}{ds} = -\mu c k^2 \delta_k + \eta_k(s), \tag{2}$$

and white Markovian FDT noise

$$\langle \eta_k(s)\eta_{k'}(s')\rangle = (2\pi)^3\delta(k+k')\,2\mu\Theta\,\delta(s-s'). \tag{3}$$

This gives the transfer kernel

$$T_R(k) = \exp[-\mu c \Delta s\, k^2] = \exp[-(k/k_c)^2], \qquad k_c = (\mu c \Delta s)^{-1/2}. \tag{4}$$

Thus the shape parameter is fixed to $p = 2$ in the v1.6 mainline toy closure. The regenerated component becomes

$$P_\xi(k) = A_F\,\frac{1-\exp[-2(k/k_c)^2]}{(k/k_c)^2}, \tag{5}$$

with finite low-k limit

$$\lim_{k\to 0} P_\xi(k) = 2A_F. \tag{6}$$

The former free power-law axis n_ξ is therefore removed from the mainline closure.

Restricted audit result

The v1.6-E audit executed only the preregistered 140-cell grid:

Axis	Values
n_s proxy	$\{0.9, 0.965, 1.0, 1.1\}$
k_c	$\{0.03, 0.1, 0.3, 1.0, 3.0\}$
A_F	$\{0.01, 0.03, 0.1, 0.3, 1.0, 3.0, 10.0\}$
p	fixed to 2
n_ξ	removed

The deterministic classification counts were:

Classification	Count
coherent_toy	25
weak	52
critical	28
failed	35
Total	140

The coherent toy fraction is $25/140 \simeq 17.86\%$. This fraction must not be compared as a physical improvement against earlier scans with different parameter axes. The meaningful improvement is structural: the free n_ξ axis is removed and $P_\xi(k)$ is tied to a conditional FDT toy closure.

Interpretation

The v1.6 branch improves the internal discipline of the perturbative restart toy model:

- $p = 2$ is no longer a free mainline parameter; it is the kernel shape produced by local isotropic Gaussian/diffusive smoothing.
- $P_\xi(k)$ is no longer a generic power law in the mainline; it follows a fixed conditional FDT shape once A_F and k_c are specified.
- The coherent toy region remains non-empty under the restricted closure.

However, the route is still not physically closed:

- $k_c = (\mu c \Delta s)^{-1/2}$ is a composite bath scale, not a cosmologically derived number.
- A_F, Θ, μ, c, and Δs are not derived from SCT cosmology.
- The perturbation variable δ is not yet a gauge-invariant cosmological observable.
- The bath/free-energy model is auxiliary and conditional.

Therefore the correct status is: conditional toy restart non-empty; physical restart not established.

Publication update

If v1.6-F is included, the recommended public release version is 1.6.0. The upload set should include the patched main English and Italian papers, the v1.5-F toy kernel addendum, this v1.6-F addendum, the v1.4-F reproducibility supplement, and the v1.6-E conditional FDT audit package. No text should describe the result as a physical cosmological prediction.

Decision

SCT_v1_6_F_CONDITIONAL_FDT_CLOSURE_INTERPRETATION_AND_PUBLICATION_UPDATE_DONE;
V1_6_CONDITIONAL_FDT_BRANCH_CLOSED_AS_TOY_REFINEMENT;
MODEL_A_CRITICAL_MASSLESS_FDT_CLOSURE_INTERPRETED;
P_EQUALS_2_AND_P_XI_FDT_SHAPE_RETAINED_AS_CONDITIONAL_TOY_MAINLINE;
K_C_AND_A_F_REMAIN_NON_DERIVED_COMPOSITE_PARAMETERS;
COHERENT_TOY_REGION_NONEMPTY_BUT_NOT_PHYSICAL_CLAIM;
ZENODO_UPLOAD_SET_UPDATED_TO_VERSION_1_6_0_IF_INCLUDED;
NO_NEW_RESTART_SCAN;
NO_PARAMETER_TUNING;
NO_MICROPHYSICAL_DERIVATION_CLAIM;
NO_PHYSICAL_COSMOLOGY_CLAIM;
NO_OBSERVATIONAL_CLAIM;
NOT_PEER_REVIEWED.